25901055 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION November 2025

First SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCSEG100 - PSYCHOACOUSTICS

(2025 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer any **20** questions

Each question carries **1** marks

1. Cerumen is produced in which part of the ear? [K] / [CO1]

   a). Middle ear b). Inner ear

   c). Ear canal d). Cochlea

2. ITD cues are most useful for which frequency range? [U] / [CO2]

   a). Low frequencies b). Very high frequencies

   c). Ultrasound d). No frequency dependence

3. Leitmotif in film scoring means: [U] / [CO4]

a). Maintain a constant emotional state
b). Recurring musical theme tied to a character or idea
c). Draw the audience's attention and increase drama
d). Compression technique

4. The sone scale is meant to be: [K] / [CO2]

a). Linear with loudness perception
b). Logarithmic to frequency
c). Related to taste
d). A time unit

5. Cultural context can alter the perceived emotion of a piece of music because of: [U] / [CO3]

a). Universal acoustics only
b). Learned associations
c). Temperature
d). Eye color

6. A lossless format means: [K] / [CO4]

a). Original audio can be perfectly reconstructed
b). Some audio data is discarded
c). Audio is converted to images
d). File is unplayable

7. Which structure helps with balance and is part of the inner ear? [U] / [CO1]

a). Cochlea
b). Semicircular canals
c). Malleus
d). Eustachian tube

8. ILD cues are most effective for: [U] / [CO2]

a). Low-frequency sounds
b). High-frequency sounds
c). Inaudible sounds
d). Sub-bass frequencies

9. Research into spatial audio often examines: [K] / [CO4]

a). HRTF personalization and rendering accuracy
b). Loudness safety
c). Music trends
d). Colour theory

10. If a tone is 440 Hz, one octave above is: [U] / [CO2]

a). 220 Hz b). 880 Hz

c). 660 Hz d). 4400 Hz

11. Claims about binaural beats improving concentration or mood are: [U] / [CO3]

a). Strongly proven b). Mixed and need more research

c). Universally false d). Related to taste

12. To reduce reverberation you would: [U] / [CO4]

a). Add reflective surfaces b). Introduce absorptive materials

c). Increase volume d). Decrease temperature

13. The brain region primarily involved in early auditory cortical processing is: [U] / [CO2]

a). Visual cortex b). Auditory cortex (temporal lobe)

c). Motor cortex d). Occipital lobe

14. Temporal masking includes both forward and: [K] / [CO2]

a). Backward masking b). Lateral masking

c). Visual masking d). Gustatory masking

15. If a high-frequency component is significantly masked, the encoder might: [U] / [CO4]

a). Discard or quantize it heavily b). Increase its amplitude

c). Add a low-frequency tone d). Change its phase

16. Critical bands describe frequency ranges within which: [K] / [CO2]

a). Spectral components interact strongly (masking) b). Colour perception occurs

c). Loudness is constant d). Sound diffracts

17. Music therapy often targets limbic responses to: [U] / [CO3]

a). Improve digestion b). Regulate mood and memory

c). Increase height d). Alter eye color

18. Binaural rendering for headphones uses: [K] / [CO4]

a). Mono audio only b). HRTFs to simulate direction

c). Lossless compression only d). Visual cues only

19. Timbre differentiates two sounds that have the same pitch and loudness by their: [K] / [CO2]

a). Duration b). Harmonic content and spectral envelope

c). Speed d). Temperature

20. Sequential grouping helps in perceiving: [K] / [CO2]

a). A sequence of notes as a melody b). Time sequences

c). Colour gradients d). Temperature trends

21. Skilled musicians typically have: [U] / [CO2]

a). Better time discrimination thresholds b). Worse pitch perception

c). Poorer memory d). No difference from non-musicians

22. If two tones have equal SPL but one sounds louder, what might explain it? [U] / [CO2]

a). Different frequency content and ear sensitivity b). Different colour

c). Different brightness d). Different weight

[1x20 = 20]

Part B

Multiple Choice Questions

Answer any **15** questions

Each question carries **2** marks

23. A lesion in the cochlear nerve would most directly produce which deficit? [U] / [CO1]

a). Conductive loss with normal bone conduction

b). Sensorineural loss with elevated air and bone thresholds

c). Purely vestibular symptoms

d). Increased cerumen production

24. High cognitive load during listening tasks commonly reduces: [U] / [CO3]

a). Speech intelligibility and working memory performance

b). Eye colour

c). Hearing thresholds only

d). None of the above

25. When designing an alarm for a noisy factory floor, it must be: [A] / [CO4]

a). Quiet, low-pitched, and easily masked

b). Highly salient with an irregular temporal pattern

c). Identical to the background noise

d). Purely visual

26. Why is temporal resolution important for speech perception? [U] / [CO2]

a). It isn't relevant.

b). It allows discrimination of rapid consonant transitions and voice-onset times.

c). It only matters for music rhythm, not speech.

d). None of the above

27. When simulating a sound at a given azimuth in headphones, convolving the source with HRTFs for left and right ear ensures: [U] / [CO2]

a). Mono playback

b). Proper interaural spectral and level differences to create directional cues

c). Lossless compression

d). Elimination of reverberation

28. Which codec family is historically associated with perceptual lossy audio coding? [U] / [CO4]

a). ZIP

b). MPEG/MP3/AAC family

c). PNG

d). TIFF

29. A classroom amplification system mainly helps students with mild to moderate hearing loss by: [U] / [CO1]

a). Increasing classroom temperature

b). Improving signal-to-noise ratio at the ear

c). Changing visual aids

d). Removing echoes entirely

30. Cross-modal interactions in the brain suggest: [K] / [CO3]

a). Auditory and visual signals are processed in isolation always

b). Sensory systems frequently interact and influence each other's perception

c). Only one sense can be active at a time

d). Vision is irrelevant to sound perception

31. Assertion (A): The most effective noise control strategy is always simply reducing the overall dB SPL level. [K] / [CO4]

Reason (R): Psychoacoustic noise control often focuses on eliminating annoying tonal components (whines/hums) and temporal impulsiveness, which significantly reduces perceived annoyance even if the total SPL remains similar.

a). Both A and R are true, and R is the correct explanation for A

b). Both A and R are true, but R is NOT the correct explanation for A

c). A is true, but R is false

d). A is false, but R is true

32. Which audiometric pattern indicates a conductive component? [U] / [CO1]

a). Air conduction thresholds equal bone conduction thresholds (no gap).

b). Air conduction thresholds are worse than bone conduction by a significant gap.

c). Only bone conduction thresholds are elevated.

d). Inverted tympanogram always correlates with sensorineural loss.

33. In a reverberant room, which strategy might improve source localization? [U] / [CO2]

a). Close both eyes

b). Use short direct-to-reverberant ratio cues and head movements

c). Increase background music volume

d). Rely solely on visual cues

34. Why is low-latency audio processing crucial in games? [U] / [CO4]

a). To ensure sound events coincide with player actions for believable interactivity and feedback

b). To reduce file size

c). To increase reverberation

d). To save CPU at the cost of responsiveness

35. Ototoxic medications (e.g., aminoglycosides) typically cause: [K] / [CO1]

a). Conductive loss reversible by earwax removal

b). Permanent sensorineural hearing loss via hair cell damage

c). Improved spatial hearing

d). Solely vestibular hyperactivity

36. How are Shepard tones constructed? [U] / [CO3]

a). By mixing multiple sine waves that are multiples of the fundamental.

b). By superimposing complex tones separated by octaves, whose amplitude envelopes overlap and fade in/out smoothly.

c). By removing all spectral content.

d). By using only bone conduction.

37. Assertion (A): Directional microphone arrays in hearing aids reduce noise by physically blocking sound waves. [U] / [CO4]

Reason (R): Directional microphone arrays improve the Signal-to-Noise Ratio (SNR) by focusing their sensitivity toward a target speaker and spatially nulling sounds from other directions.

a). Both A and R are true, and R is the correct explanation for A

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38. Assertion (A): Inner Hair Cells (IHCs) are the cochlear transducers, converting mechanical motion to neural signals. [U] / [CO1]

Reason (R): IHCs release neurotransmitters when their stereocilia are bent by the shearing motion between the tectorial and basilar membranes.

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39. The nucleus accumbens is primarily associated with: [K] / [CO3]

a). Motor reflexes

b). Reward and pleasure responses (e.g., chills from music)

c). Visual acuity

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[2x15 = 30]

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